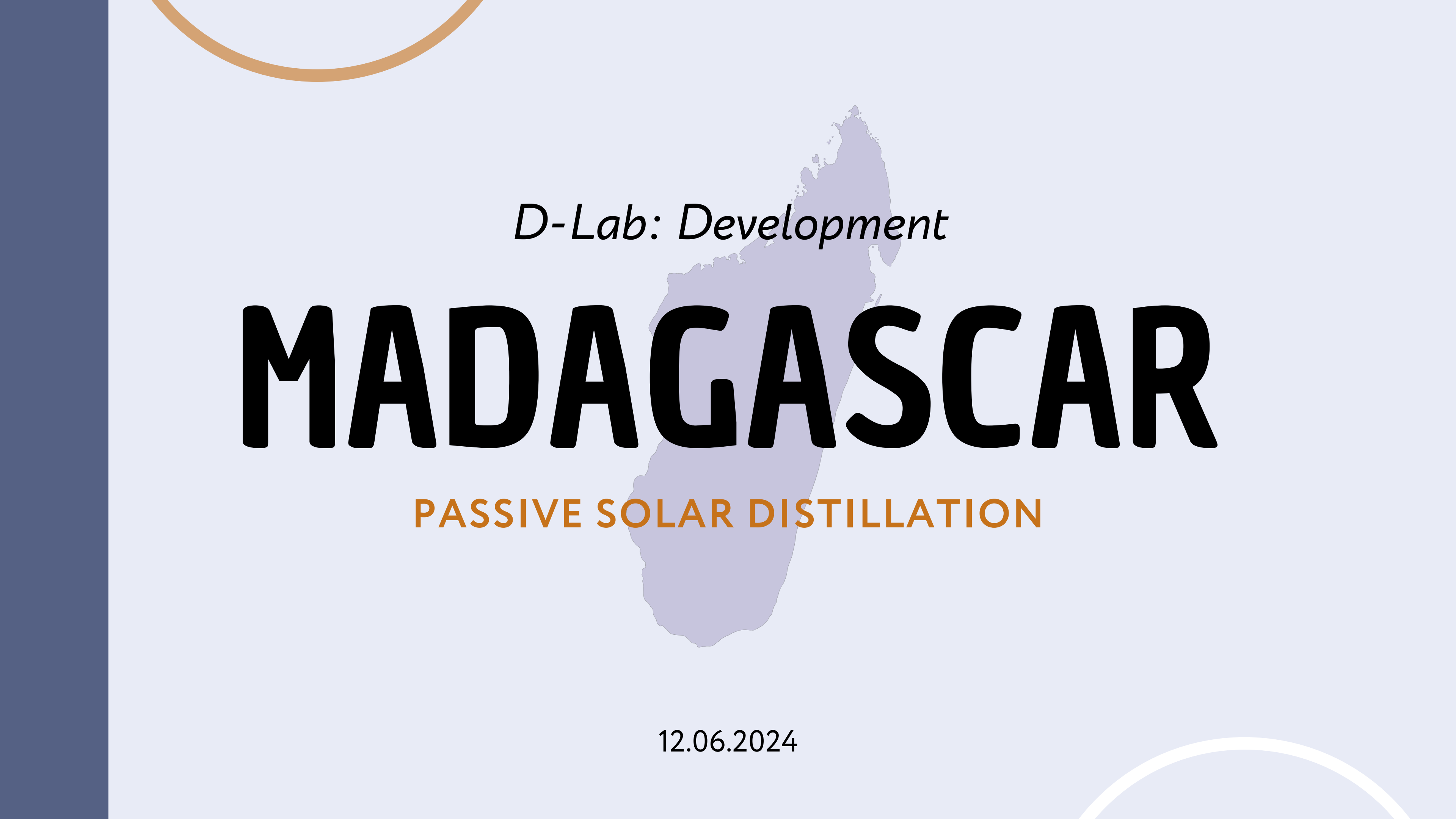




D-Lab: Development

MADAGASCAR

PASSIVE SOLAR DISTILLATION

12.06.2024

MIT Team



**JUSTIN
COLE**



**RACHEL
MOHAMMED**



**LEEBAN
MORGAN**



**LINDSAY
REYES**



Our Partners



1

**KARL
ZIMMERMAN
PROJECT LEAD**

D-Lab Alumni; Environmental Engineer at University of British Columbia; Worked on the Pano Rano project for the past 3 years!

2

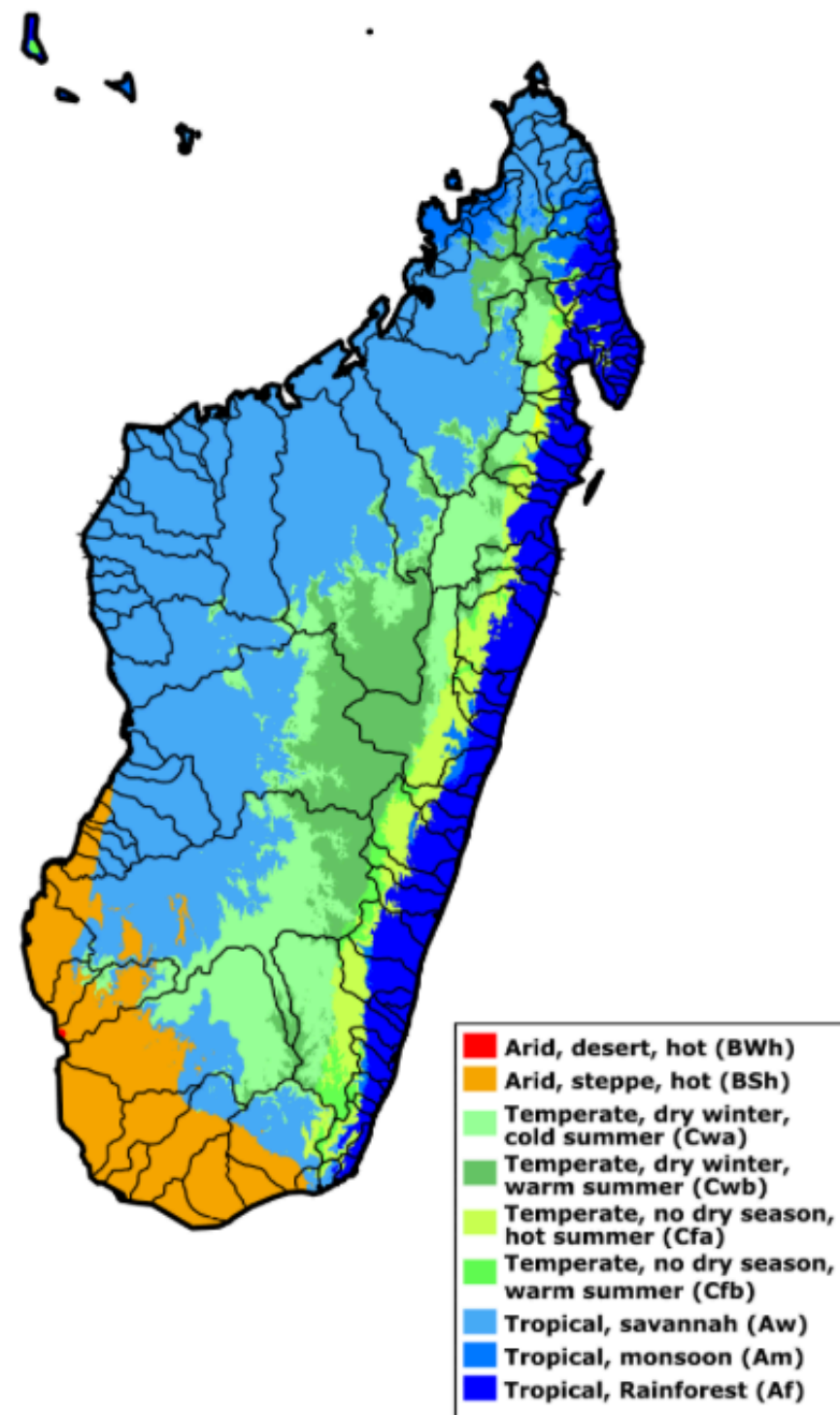
**HARRY
CHAPLIN
TATIRANO SOCIAL
ENTERPRISE**

Local nonprofit based in Fort Dauphin dedicated to providing clean water via rainwater solutions

3

**ANGELOS
ANDRIAMANAMPISOA
ÉCOLE SUPÉRIEURE POLYTECHNIQUE
D'ANTANANARIVO (ESPA)**

University Researchers:
Manampy and **Santa**
Working in tandem on an open system distiller



Location

Androy, Madagascar

- Southernmost region
- Highest poverty rate on the island
- Hot, arid climate
- <400mm of rainfall annually

Antananarivo
Fort-Dauphin
Ambovombe

.....
.....
.....

Capital of Madagascar
Capital of Androy Region
Testing Site in Androy

Passive Solar Distillers

1

Energy from the sun is trapped inside

2

Salty water evaporates in the box

3

Water vapor condenses on the glass pane

4

Clean water collects at the bottom!



Motivation

- Freshwater is scarce
- 70% of people lack basic drinking water
- Half of the region requires water, sanitation, and hygiene assistance
- Locals often drink salt water from riverbank and wells
- Or drink potentially contaminated groundwater

Theory of Change

IMPACT

OUTCOME

OUTPUT

INPUT

Solar distiller

Communities
install
distillers with
Tatirano/ESPA

Greater
supply of
drinkable
sweet water

Cleaner water for
agriculture and drinking,
improving health

Previous Work

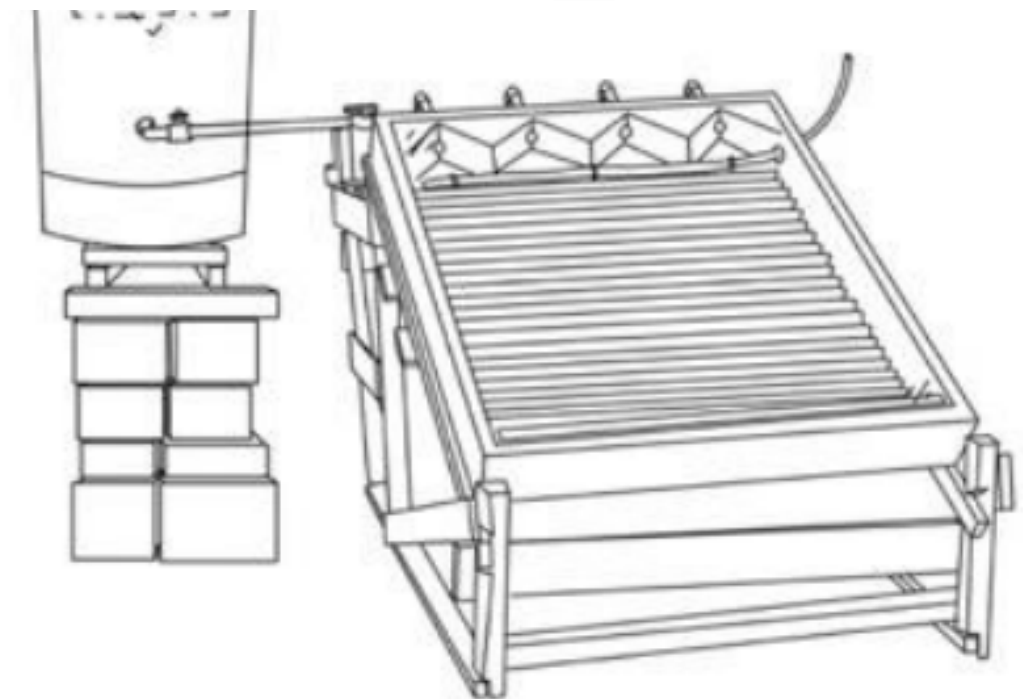
THE PANO RANO DISTILLER

Concrete-cast box with a glass sheet on top

- Open Design: Condensation occurs outside of the evaporation chamber
- Closed Design: Both evaporation and condensation occur in same chamber

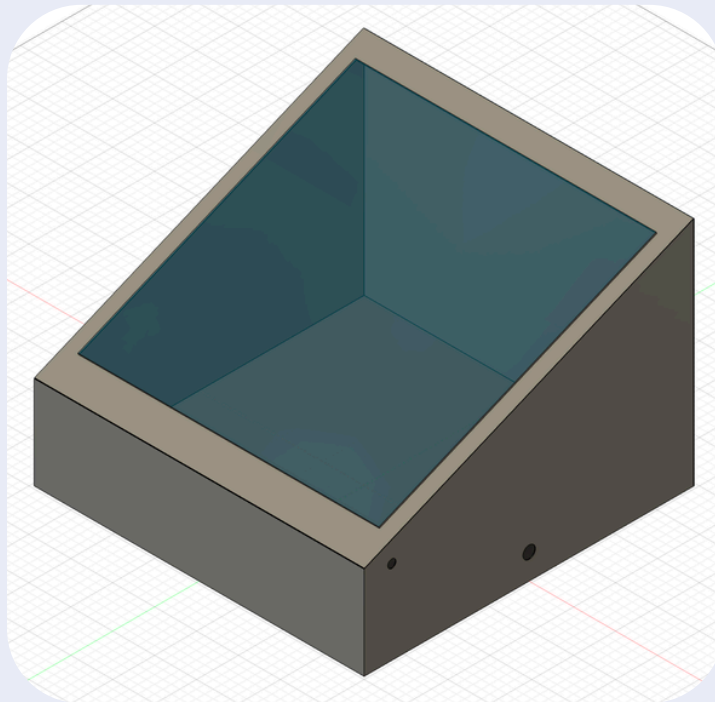
CHALLENGES

- Concrete weight and heat absorption
- Limited water capacity
- Condenser complexity



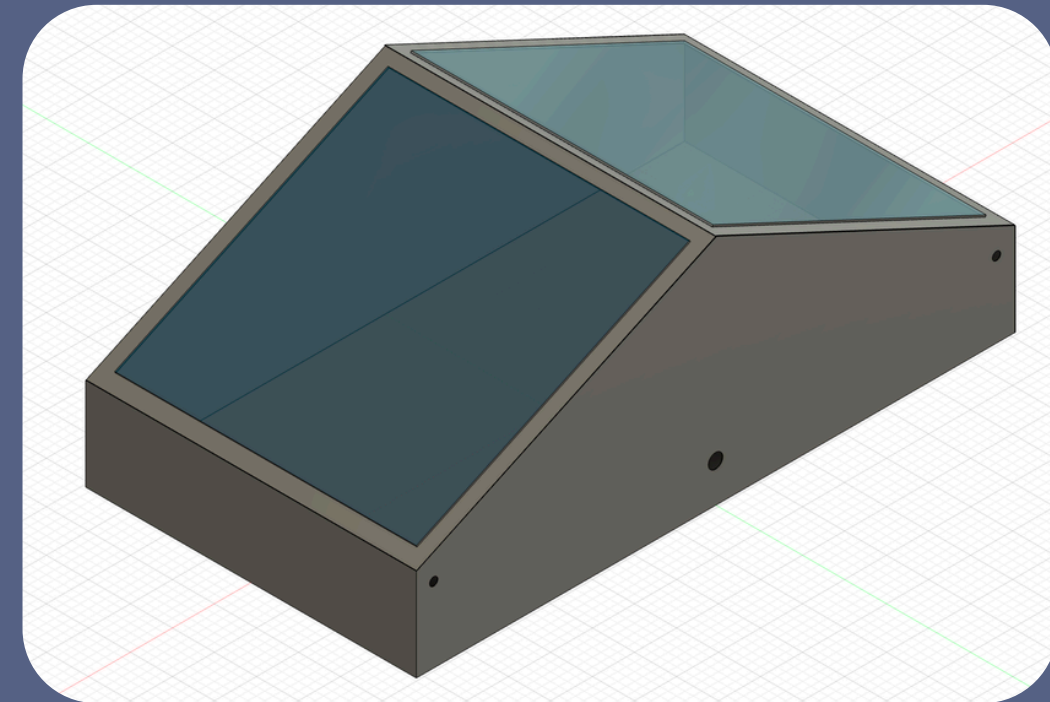
Distiller Designs

SINGLE SLOPE

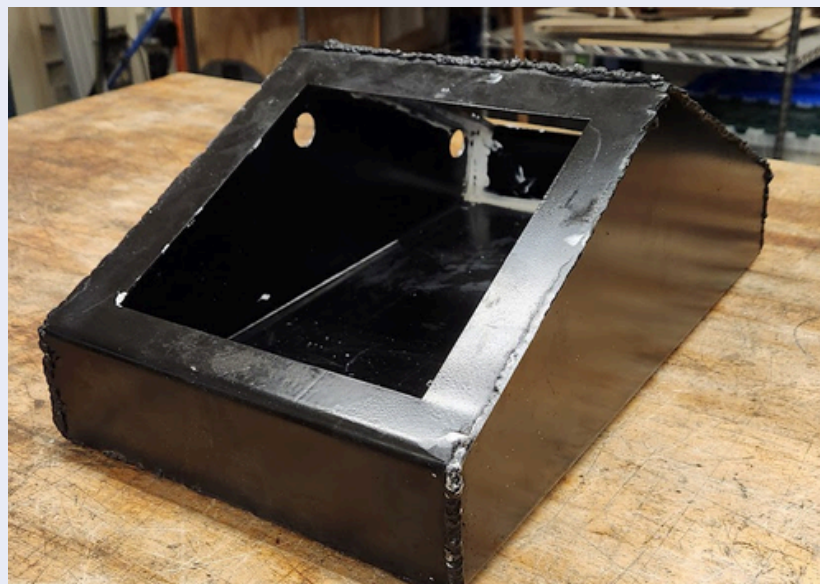


- Control Design that mimic previous Pano Rano characteristics
- Reduced material consumption

DOUBLE SLOPE



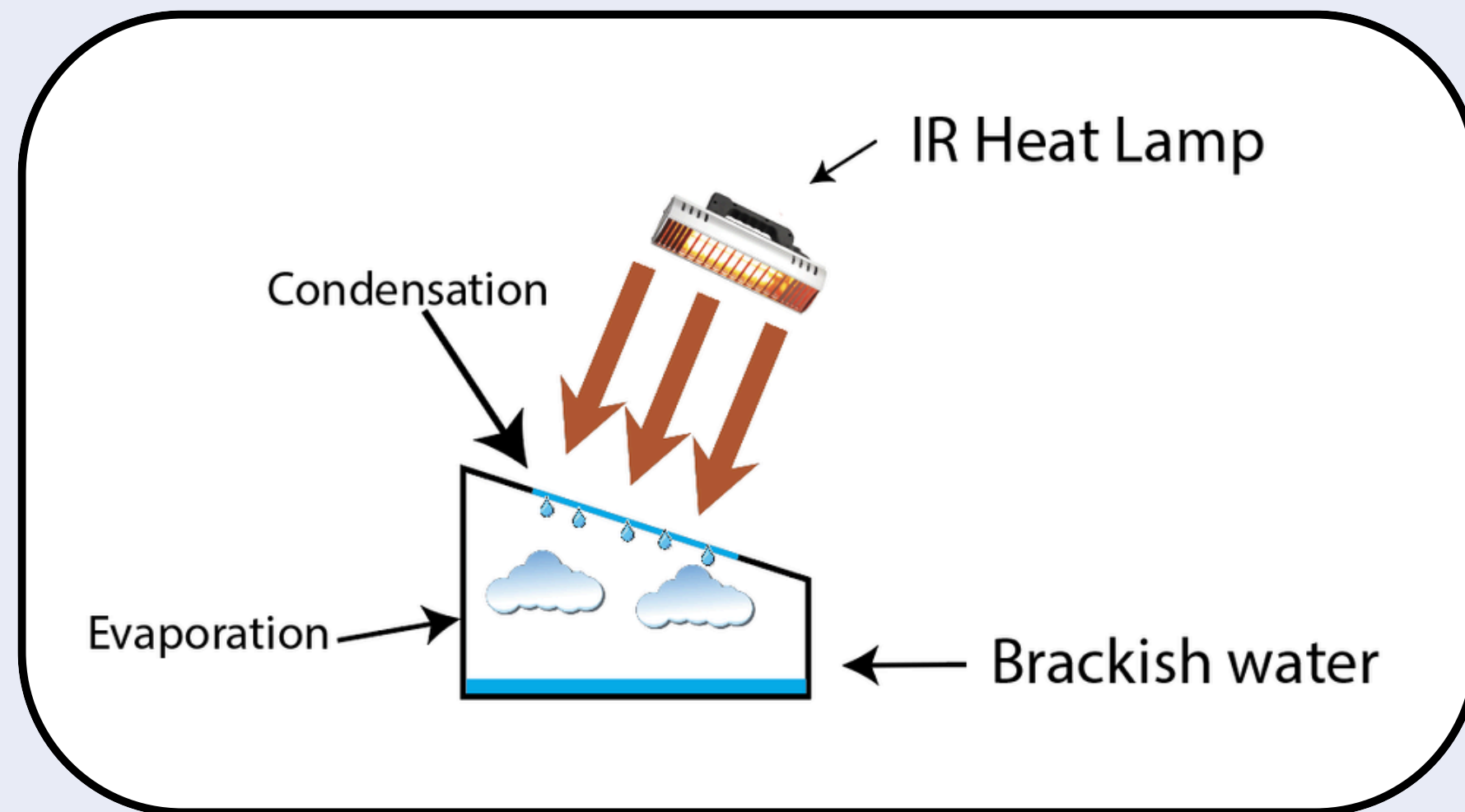
- Double the tank capacity and evaporation space
- One cooler roof pane to increase condensation



Construction Process

- Source materials (steel, glass, rubber, etc)
- Laser cut/waterjet cut steel
- Bend and weld steel to form boxes
- Paint boxes black, drill holes, and caulk to seal
- Install glass, gaskets, internal piping, and DHT-11 sensors
- Produce sweetwater! :)

Testing Plan + Data Collection



SOLAR SIMULATION

- Using a handheld infrared 1000 W lamp to simulate sunlight in Androy
- Irradiance sensor to measure sensor

SUCCESS METRICS

- Humidity data gathered from DHT11 sensor
- Mass of condensed water inside the box

IAP Fieldwork Goals

Scale Up Model

MIT team created a prototype to test during the semester. During IAP, we want to see how an upscaled version would work in the field.

Compare Open vs Closed System

Closed system: evaporating and condensing happens in the same chamber. ESPA is working on the open system: evaporating and condensing occurs in separate chambers

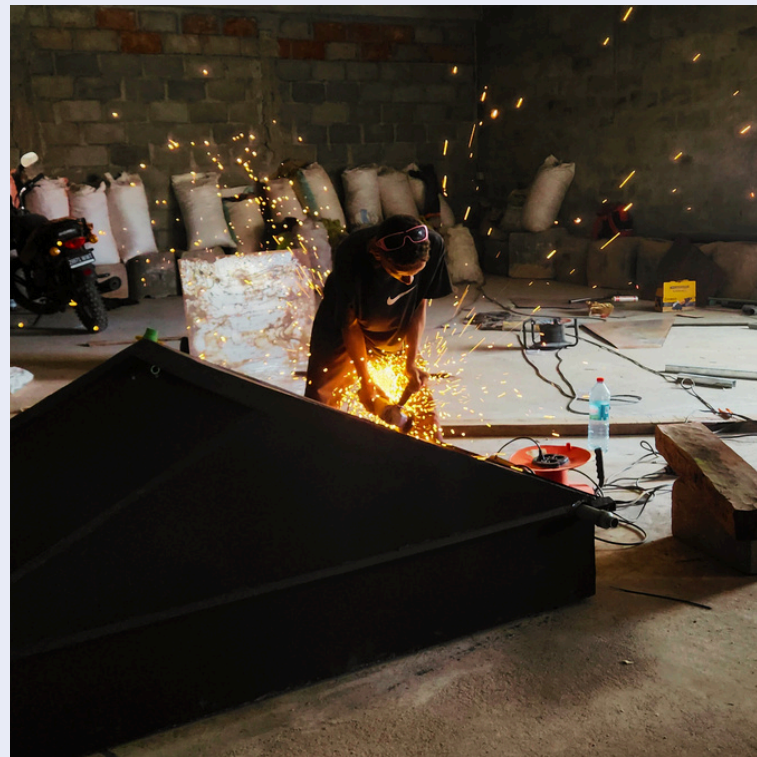
Produce $> 5L/m^2/day$

Tatirano's overall goal!

Closer to meeting WHO's standard of 50-100L/person/day to meet basic needs

Repeatable Construction

So every community can have access to safe and clean water!



Construction Process

- Source materials (steel, glass, rubber, etc) from the market
- Bend and weld steel to form boxes
- Paint boxes black, drill holes, and caulk to seal
- Install glass, gaskets, internal piping, and DHT-11 sensors
- Produce sweetwater! :)

Testing Plan + Data Collection



VARIABLES

- Orientation to Sun Trajectory
- Water Level
- Shading

DATA COLLECTED

- Condensed water collected in output bottles
- Measured temperature and humidity on each slope compared to ambient
- Collected from 7am → 5pm

Experimental Findings

MATERIALS

**Conductors lose
15-20% of
potential heat**

Steel double slope had an internal temperature of ~55 degrees celsius, while concrete pano rano reached ~70 degrees

GEOMETRY

**Double-Slope
increases
condensation**

Two slopes parallel to the sun trajectory creates a “warm” and “cool” side, making it easier for water to condense on the cool side, which increases the condensation rate for a given internal temperature

EFFICIENCY

**Double Slope and
Pano Rano had
similar outputs**

During testing, the 2025 Double Slope Distiller and 2024 Pano Ranos produced similar literage outputs (normalized)

Global Significance

About TWO BILLION people worldwide don't have access to safe drinking water today

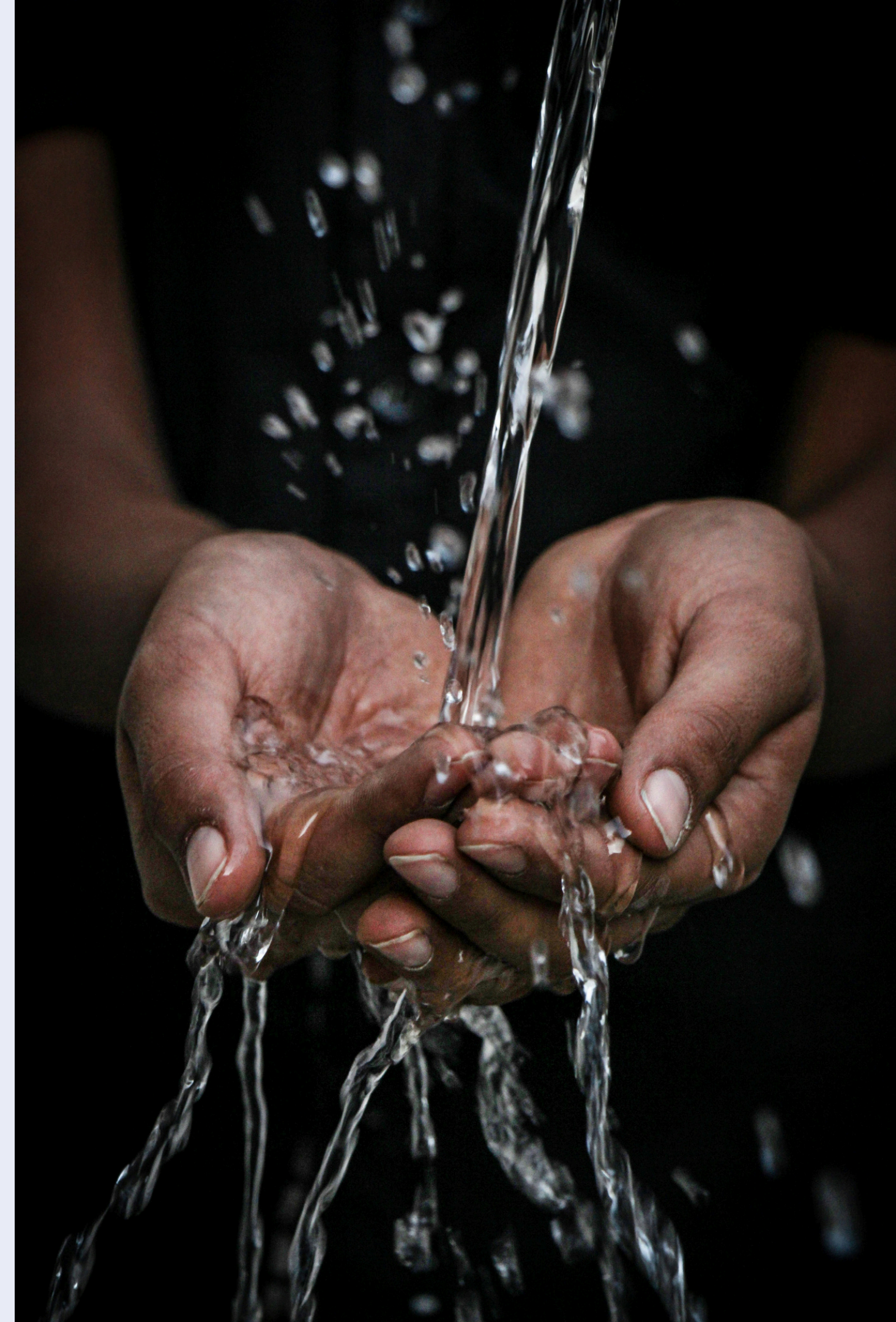
- *SDG Report 2022*

Roughly HALF of the world's population is experiencing severe water scarcity for at least part of the year

- *Intergovernmental Panel on Climate Change*

These numbers are expected to INCREASE, exacerbated by climate change and population growth

- *World Meteorological Organization*





Thanks :))))))

